

TUTORIAL 4

4) Write a C++ program to implement all arithmetic operations using separate member functions.

INPUT:

```
#include <iostream>
using namespace std;
```

```
class Arithmetic
```

```
{
```

```
private:
```

```
    int a, b;
```

```
public:
```

```
    void getData()
```

```
    {
```

```
        cout << "Enter two numbers: ";
```

```
        cin >> a >> b;
```

```
    }
```

```
    void addition()
```

```
    {
```

```
        cout << "Addition = " << a + b << endl;
```

```
    }
```

```
    void subtraction()
```

```
    {
```

```
        cout << "Subtraction = " << a - b << endl;
```

```
    }
```

```
    void multiplication()
```

```
    {
```

```
        cout << "Multiplication = " << a * b << endl;
```

```
    }
```

```
    void division()
```

```
    {
```

```
        if (b != 0)
```

```
            cout << "Division = " << (float)a / b << endl;
```

```
        else
```

```
            cout << "Division by zero is not possible." << endl;
```

```
    }
```

```
void modulus()
{
    if (b != 0)
        cout << "Modulus = " << a % b << endl;
    else
        cout << "Modulus by zero is not possible." << endl;
}

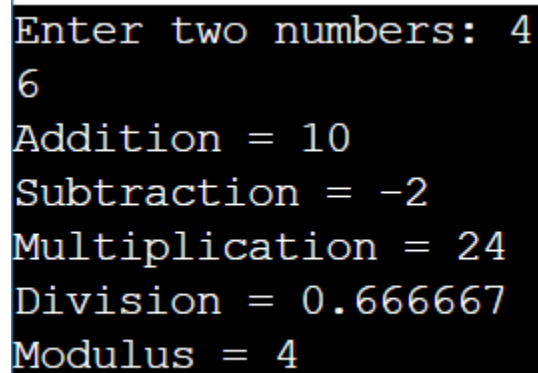
};

int main()
{
    Arithmetic obj;

    obj.getData();
    obj.addition();
    obj.subtraction();
    obj.multiplication();
    obj.division();
    obj.modulus();

    return 0;
}
```

OUTPUT:

A screenshot of a terminal window with a black background and yellow text. The output shows the results of a program where two numbers, 4 and 6, are entered. The program then displays the results of five operations: Addition (10), Subtraction (-2), Multiplication (24), Division (0.666667), and Modulus (4).

```
Enter two numbers: 4
6
Addition = 10
Subtraction = -2
Multiplication = 24
Division = 0.666667
Modulus = 4
```